

## 软件系统设计 2015 年考题整理

1. Where do software architecture come from? List five possible sources of software architecture.

软件架构的来源是什么？列出五个可能的软件架构来源。

2. What distinguishes an architecture for a software product line from an architecture for a single product?

软件产品线的架构与单个产品的架构有何区别？

3. How to model quality attribute scenarios? Graphically model two quality attributes in “stimulus-response” format: availability and performance.

如何对质量属性场景进行建模？以“刺激-响应”的格式以图形方式对可用性和性能这两个质量属性进行建模。

4. Describe the relationships between architectural patterns and tactics in architecture design. Give name of any four tactics that can be used during architecture design and describe the purpose of each of them.

请描述架构模式与架构设计中的策略之间的关系。提供四种在架构设计中可以使用的策略名称，并描述每种策略的目的。

5. Briefly describe the general activities involved in a software architecture process.

简要描述软件架构过程中涉及的一般活动。

6. Map each of the following questions (on the left) with the architectural style/view (on the right) that addresses the question. List four views of each category of style.

- |                                                                                          |                              |
|------------------------------------------------------------------------------------------|------------------------------|
| • How it is structured as a set of implementation units?                                 | • Component-connector styles |
| • How it is structured as a set of elements that have runtime behavior and interactions? | • Allocation styles          |
| • How it relates to non-software structures in its environment?                          | • Module styles              |

将以下问题（左侧）与解决该问题的架构风格/视图（右侧）进行对应。列出每个风格类别的四个视图。

- |                               |            |
|-------------------------------|------------|
| • 它如何作为一组实现单元进行结构化？           | • 组件-连接器风格 |
| • 它如何作为具有运行时行为和相互作用的元素集进行结构化？ | • 分配风格     |
| • 它与其环境中的非软件结构之间的关系如何？        | • 模块风格     |

7. Explain the context, benefits and limitations of Broker Architecture Pattern.

解释代理架构模式的上下文、好处和局限性。

8. Why should a software architecture be documented using different views? Give the names and purposes of four example views.

为什么软件系统架构需要使用不同视图来文档化？给出四个示例视图的名称和目的。

9. Briefly describe the fundamental principles of Service Oriented Architecture(SOA) and discuss the impact of SO on quality attributes like interoperability, scalability and security.

简要描述面向服务的架构 (SOA) 的基本原则, 并讨论 SO 对互操作性、可扩展性和安全性等质量属性的影响。

**10.** Describe the outputs generated from each phase of Architecture Tradeoff Analysis Method(ATAM) process.

描述架构权衡分析方法 (ATAM) 过程的每个阶段生成的输出。

**11.** How do software product lines and model driven architecture archive a high reusability? Compare and discuss the commonalities and differences between these two architectural approaches from various perspectives.

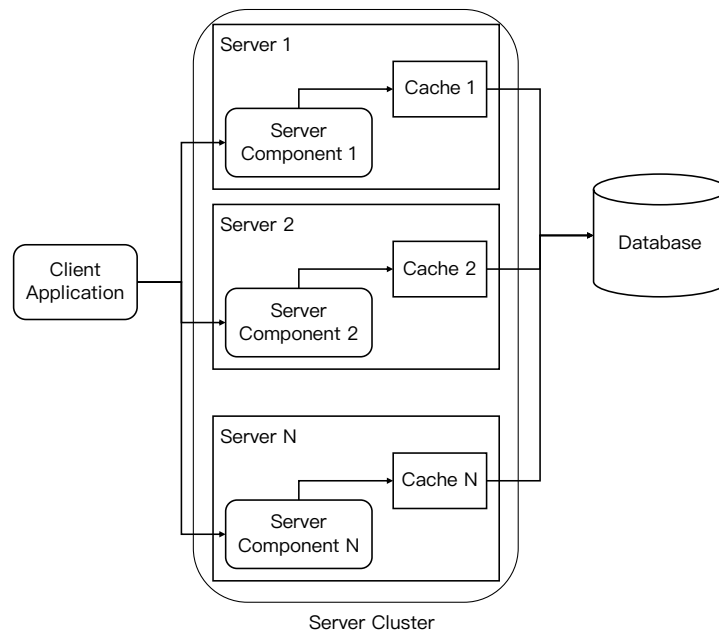
软件产品线和模型驱动架构如何实现高可重用性? 从多个角度比较和讨论这两种架构方法的共同点和差异。

**12.** Distributed Cache Updates. Figure 1 below shows a distributed application with caches to improve its run-time performance. An online customer service system provides its clients access to their Bill Statement Information(BSI). The database stores BSI for clients. During a session, a client may require the BSI several times. In order to improve the performance, the BSI information for a particular client is cached. If the cache is loaded, then the BSI can be directly returned back to the client without access to the database. There are N servers clustered to serve the requests from clients. The server is load-balanced, and each time one server is scheduled for one client request. All the caches on N servers need to synchronize their states. If the client request changes the states of one cache, there are two requirements:

1. The changes are committed to the database;
2. All other cache states must be invalidated and they should reload the states from the database.

The current design in Figure 1 cannot meet the second requirement. Some important components are missing. Consider to apply architectural pattern(s) and tactic(s) to the design that can meet the second requirement, and complete the following questions:

1. Identify extra component(s) that is (are) necessary for this design to meet second requirement. List each component name and its major functionalities.
2. Draw sequence diagram to illustrate the communication between your devised components and original components in Figure 1.
3. Assume that the servers are heterogeneous, and the protocols to access caches on different servers vary. For example, Server 1 supports Java RMI invocations to cache 1, Server 2 supports http JSP requests to cache 2, and server 3 supports SOAP requests to cache 3. It is further assumed that the cache interfaces are not consistent to each other due to the evolution of system development and legacy problems. Explain how the concept of connectors or web services can be applied to provide a generic invocation to caches? You can choose either connector or web services for discussion.



分布式缓存更新。下图 1 展示了一个带有缓存的分布式应用程序，以提高其运行时性能。在线客户服务系统为客户提供访问其账单信息 (BSI) 的功能。数据库存储着客户的 BSI 信息。在会话期间，客户可能多次需要访问 BSI。为了提高性能，特定客户的 BSI 信息被缓存起来。如果缓存已加载，则可以直接将 BSI 返回给客户而无需访问数据库。有  $N$  个服务器集群用于为客户请求提供服务。服务器进行负载均衡，并且每次将一个服务器调度给一个客户请求。 $N$  个服务器上的所有缓存需要同步它们的状态。如果客户请求更改了一个缓存的状态，则有两个要求：

1. 更改必须提交到数据库；
2. 所有其他缓存状态必须失效，并且它们应从数据库重新加载状态。

图 1 中的当前设计无法满足第二个要求。缺少一些重要组件。请考虑应用架构模式和策略来满足第二个要求，并完成以下问题：

- (1). 确定此设计要满足第二个要求所需的额外组件。列出每个组件的名称和其主要功能。
- (2). 绘制序列图以说明您设计的组件与图 1 中的原始组件之间的通信。
- (3). 假设服务器是异构的，并且访问不同服务器上的缓存的协议各不相同。例如，服务器 1 支持对缓存 1 的 Java RMI 调用，服务器 2 支持对缓存 2 的 http JSP 请求，服务器 3 支持对缓存 3 的 SOAP 请求。进一步假设由于系统开发的演进和遗留问题，缓存接口不一致。请解释如何应用连接器或 Web 服务的概念来提供对缓存的通用调用？您可以选择连接器或 Web 服务进行讨论。